

Federated belief sharing from private observations to claim-bounded consensus

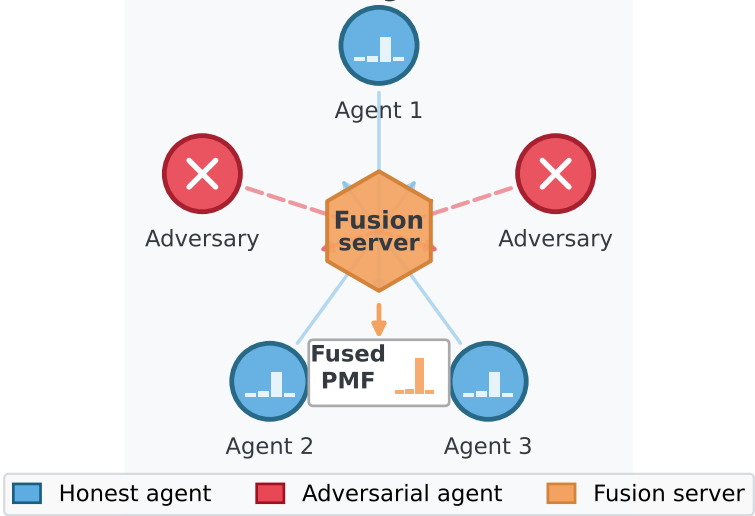
Categorical active inference + generalized Bayes

Project identity: `robust_aggregate(c=0) ≡ log_linear_pool`

Friston Eq. 7: qualified categorical message combination under shared support, posterior-log potentials, and fixed weights

1

Federated Belief Sharing Under Contamination



2

Consensus after contamination

Same broadcasts, different fusion rule

Naive log-linear pool

confident wrong broadcasts pull the product



39%

+11pp true-state mass

Heuristic reweighting

configured divergence reweighting before fusion



50%

Deterministic schematic: 2/5 agents contaminated

Variational server: objective-backed fusion is a separate conservative rule

3

Three robustness axes — related routes, non-transferable guarantees

CLIENT-SIDE FEDGVI

source / β generalized-Bayes update
source-conditional bounded-influence result

SERVER HEURISTIC

`robust_aggregate` reweighting
conditional accuracy; recovery limit only

VARIATIONAL SERVER

`variational_aggregate` and $F(q,a)$
objective-backed raw-weight control; conservative